

Model Question Paper set-I**SYNCHRONOUS MACHINES AND FHP MOTORS***Time: 3 Hour**Max.Marks: 75***PART A**

- I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

1	Name the alternator which has large axial length and small diameter.	M1.01	U
2	Under purely inductive load the armature reaction has _____ effect in alternator.	M1.03	R
3	Load angle is the phase difference between ____ and ____.	M1.04	R
4	Identify the voltage regulation method which is known as the pessimistic method used for alternators.	M2.01	U
5	Name any two methods to synchronize two alternators.	M2.04	R
6	Write the necessary conditions to operate synchronous motors as synchronous condensers.	M3.04	U
7	The V curve of a synchronous motor shows the relationship between.	M3.02	R
8	Describe the method to eliminate the hunting effect on synchronous motors.	M3.02	U
9	List out the classification of stepper motor based on operation.	M4.03	R

PART B

II. Answer any **eight** questions from the following, each question carries 3 marks.

1	Calculate distribution factor of a three phase distributed winding of alternator. The alternator has 2 poles and 18 slots.	M1.02	U
2	List out any three advantages of stationary armature over rotating armature.	M1.01	R
3	Describe the ZPF characteristics of a synchronous generator.	M2.01	U
4	Explain the necessary conditions to be satisfied for parallel operation of alternators.	M2.03	R
5	Describe the experimental method to plot the open circuit characteristics of the alternator.	M2.01	U
6	Define the term synchronizing current in parallel operation of alternators.	M2.03	R
7	Develop the power flow diagram of a synchronous motor.	M3.02	R
8	Identify the basic features of synchronous motors.	M3.01	U
9	Describe the necessary conditions to be satisfied to self start a single phase induction motor.	M4.01	U
10	Explain why a synchronous motor is known as a doubly excited machine.	M3.01	R

PART C

Answer ALL questions. Each question carries 7 marks.

III	Find phase and line voltages of star connected three phase 6 pole alternator which runs at 1200 RPM having flux per pole of 0.1 webber sinusoidal distributed, its stator has 54 slots. 2 conductors per slot each coil short pitched by one slot.	M1.02	A
	OR		
IV	Calculate the RMS value of induced EMF per phase of a 10 pole three phase 50 Hz alternator with 2 slots per pole per phase and 4 conductor per slot. The coil span is 150 degrees. Flux per pole is 0.12 web.	M1.02	A
V	Compare the salient pole and cylindrical rotor type synchronous generators.	M1.01	R
	OR		
VI	List out the advantages of short pitched and distributed type windings in alternators.	M1.02	R
VII	From the following test results determine the voltage regulation of a 2000 V single phase alternator delivering a current of 100 A at 0.8 pf leading. The results: full load Current of 100 A is produced on short circuit by a field excitation of 2.5 A and EMF of 500 V is produced on open circuit by the same field current. The armature resistance is 0.8 ohm.	M2.01	A
	OR		
VIII	A 200 KVA 3000 V 50 Hz three phase alternator has effective armature resistance of 0.6 ohm per phase. A field current of 45 A produces a short circuit current of 240 A on short circuit and open circuit voltage of 1040 V between lines. Calculate percentage voltage regulation at 0.8 pf lag.	M2.01	A

IX	Explain the working principle of synchronous motors with the help of neat diagrams.	M3.01	U
	OR		
X	Compare three phase induction motor and synchronous motor.	M3.04	U
XI	Explain with the help of double field revolving theory that the single phase induction motor is not a self starting motor.	M4.01	U
	OR		
XII	Explain the construction and working principle of repulsion type motors.	M4.02	U
XIII	Describe the working of a servo motor with the help of block diagram.	M4.03	U
	OR		
XIV	Explain the working principle of permanent capacitor type single phase induction motor with the help of circuit diagram and vector diagram.	M4.01	U

Model Question Paper set II

SYNCHRONOUS MACHINES AND FHP MOTORS

Time: 3 Hour

Max.Marks: 75

PART A

- I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

1	Number of brushes required in a rotating field type alternator to provide DC excitation.	M1.01	R
2	Identify the type of alternator for high speed operations.	M1.01	U
3	Synchronous reactance consists of _____ and _____.	M1.04	R
4	List any two methods to find voltage regulation of alternator.	M2.01	R
5	Define the term synchronizing torque in parallel operation of alternators.	M2.04	R
6	If excitation current is increased in the synchronous motor under lagging power factor load the power factor will _____.	M3.02	U
7	List any two features of synchronous motors.	M3.01	R
8	Write the equation for synchronous speed of synchronous motor.	M3.01	U
9	Write any two applications of universal motor.	M4.04	R

PART B

II. Answer any **eight** questions from the following, each question carries 3 marks.

1	Calculate pitch factor of given alternator if the coil span of winding is 150 degree.	M1.02	U
2	Describe the effect of armature reaction under leading power factor load in alternators.	M1.03	U
3	Illustrate the nature of open circuit and short circuit characteristics of alternators.	M2.01	U
4	Draw the layout of parallel operation of two alternators in the dark lamp method.	M2.03	R
5	List out any three advantages of parallel operation of alternators.	M2.03	R
6	Describe the operation of a synchronous motor as a synchronous condenser.	M3.02	U
7	Define any three torques in synchronous motors.	M3.02	R
8	List out any three applications of synchronous motor.	M3.04	R
9	Describe the nature of capacitors used in capacitor start induction run type single phase induction motors.	M4.01	U
10	Describe how the damper winding helps the self starting of the synchronous motor.	M3.02	U

PART C

Answer ALL questions. Each question carries 7 marks.

III	Calculate speed and open circuit phase voltage of a 4 pole 3 phase 50 Hz star connected alternator with 36 slots and 30 conductor per	M1.02	A
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	slot full pitched winding. The flux per pole is 0.05 web sinusoidally distributed.														
	OR														
IV	A 3 phase 8 pole 750 rpm star connected alternator has 72 slots on armature.Each slot has 12 conductors and winding is short chorde by 2 slots.Find induced emf between lines,given flux per pole is 0.06 web.	M1.02	A												
V	Derive the equation for generated EMF in synchronous generators.	M1.02	U												
	OR														
VI	Develop the phasor diagrams of alternators under lag lead and unity power factor loads.	M1.04	U												
VII	A three phase star connected 1000 KVA,11000V alternator has rated current of 52.5A. the ac resistance per phase is 0.45 ohm.The test result are given below OC test : field current = 12.5 A voltage between line = 422 V SC test : field current = 12.5 A line current = 52.5 A Determine full load voltage regulation at 0.8 pf lead.	M2.01	A												
	OR														
VIII	The following test results are obtained on a 6600 V alternator. <table border="1"><tr><td>Open circuit voltage</td><td>3100</td><td>4900</td><td>6600</td><td>7500</td><td>8300</td></tr><tr><td>Field current</td><td>16</td><td>25</td><td>37.5</td><td>50</td><td>70</td></tr></table> A field current of 20A necessary to circulate full load current on a short circuit. Calculate full load regulation at 0.8 pf lag by MMF method.	Open circuit voltage	3100	4900	6600	7500	8300	Field current	16	25	37.5	50	70	M2.01	A
Open circuit voltage	3100	4900	6600	7500	8300										
Field current	16	25	37.5	50	70										
IX	Describe the different starting methods used in synchronous motors.	M3.02	U												

	OR		
X	Illustrate the experimental method to plot the V and inverted V curves of a synchronous motor.	M3.02	U
XI	Describe the construction and working of universal motors.	M4.02	U
	OR		
XII	Describe the construction and working of capacitor start capacitor run type induction motors.	M4.01	U
XIII	Explain the construction and working of switched reluctance motors.	M4.03	U
	OR		
XIV	Explain the construction and working of shaded pole motors.	M4.01	U